Example

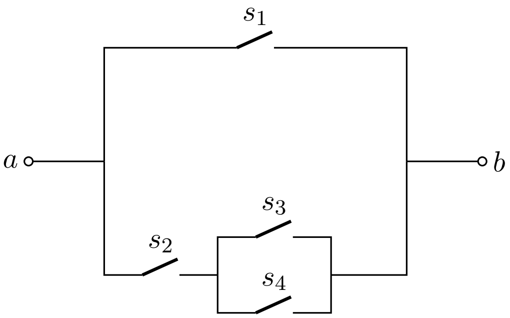
Sit 3 men and 4 women at random in a row. What is the probability that either all the men or all the women end up sitting together?

Example

A group of 3 Norwegians, 4 Swedes, and 5 Finns is seated at random around a table. Compute the probability that at least one of the three groups ends up sitting together.

Example

Consider the switching network shown in figure. It is equally likely that a switch will or will not close. Find the probability that a closed path will exist between terminals a and b.



Example

Example

Consider the experiment of tossing a fair coin repeatedly and counting the number of tosses required until the first head appears.

a. Find the sample space of the experiment.
b. Find the probability that the first head appears on the kth toss.
c. Verify that $P(S) = 1$

Example

There are 100 passengers lined up to board an airplane with 100 seats (with each seat assigned to one of the passengers). The first passenger in line crazily decides to sit in a randomly chosen seat (with all seats equally likely). Each subsequent passenger takes their assigned seat if available, and otherwise sits in a random available seat. What is the probability that the last passenger in line gets to sit in their assigned seat?